

|                                                       |                                               |
|-------------------------------------------------------|-----------------------------------------------|
| Program: <b>Diploma in Food Processing Technology</b> |                                               |
| Course Code: <b>3429</b>                              | Course Title: <b>Food Microbiology -I Lab</b> |
| Semester: <b>3</b>                                    | Credits: <b>1.5</b>                           |
| Course Category: <b>Program core</b>                  |                                               |
| Periods per week: <b>3 (L:0 T:0 P:3)</b>              | Periods per semester: <b>45</b>               |

### Course Objectives:

- The student will acquaint knowledge in preparation of cultural media
- Develop skills to analyze enumerate microorganisms using various techniques
- Identify different microorganisms or parts using various staining techniques.

### Course Pre requisites:

| Topic                               | Course Code | Course Name         | Semester |
|-------------------------------------|-------------|---------------------|----------|
| Microorganisms associated with food |             | Food microbiology I | 3        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                                                                                                               | Duration (Hours) | Cognitive Level |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Identify, explain function and preparation of common culture media in microbiology lab and sterilization                                                                  | 11               | Applying        |
| CO2             | Demonstrate proficiency in aseptic transfer of living microbes using different techniques; staining techniques, proper culture handling and enumeration of microorganisms | 17               | Applying        |
| CO3             | Demonstrate proper usage, identify the parts/functions of the microscopes and viewing microorganisms using with/ without staining techniques                              | 9                | Applying        |
| CO4             | Explain bacterial growth and environmental/physical/chemical factors that influence microbial growth                                                                      | 8                | Applying        |

## CO–PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             |     |     |     | 3   | 3   |     |     |
| CO2             |     |     |     | 3   | 3   |     |     |
| CO3             |     |     |     | 3   | 3   |     |     |
| CO4             |     |     |     | 3   | 3   |     |     |

3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped

## Course Outline

| Module Outcomes | Description                                                                                                                                                                      | Duration (Hours) | Cognitive Level |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | <b>Identify, explain function and preparation of common culture media in microbiology lab and sterilization</b>                                                                  |                  |                 |
| M1.01           | Culture media                                                                                                                                                                    | 2                | Understanding   |
| M1.02           | Preparation of nutrient broth                                                                                                                                                    | 3                | Applying        |
| M1.03           | Preparation of nutrient agar plates and nutrient agar slants                                                                                                                     | 3                | Applying        |
| M1.04           | Subculturing in slants                                                                                                                                                           | 3                | Applying        |
| CO2             | <b>Demonstrate proficiency in aseptic transfer of living microbes using different techniques; staining techniques, proper culture handling and enumeration of microorganisms</b> |                  |                 |
| M2.01           | Enumeration of heterotrophic bacterial population- Serial dilution and pour plate technique                                                                                      | 3                | Applying        |
| M2.02           | Demonstration of technique for pure culture of microorganisms-Spread plate and Streak plate method                                                                               | 6                | Applying        |
| M2.03           | Microscopic enumeration of microorganisms                                                                                                                                        | 3                | Applying        |
| M2.04           | Simple staining                                                                                                                                                                  | 3                | Applying        |

|            |                                                                                                                                                     |   |          |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---|----------|
|            | Lab Test -1                                                                                                                                         | 2 |          |
| <b>CO3</b> | <b>Demonstrate proper usage, identify the parts/functions of the microscopes and viewing microorganisms using with/ without staining techniques</b> |   |          |
| M3.01      | Negative staining of bacteria                                                                                                                       | 3 | Applying |
| M3.02      | Grams Staining method for differentiation of bacteria                                                                                               | 3 | Applying |
| M3.03      | Fungal staining                                                                                                                                     | 3 | Applying |
| <b>CO4</b> | <b>Explain bacterial growth and environmental/physical/ chemical factors that influence microbial growth</b>                                        |   |          |
| M4.01      | Bacterial Growth curve-observation and growth characteristics of bacteria                                                                           | 3 | Applying |
| M4.02      | Antibiotic sensitivity of pathogens                                                                                                                 | 3 | Applying |
|            | Lab Test-2                                                                                                                                          | 2 |          |

### Text/Reference

| <b>T/R</b> | <b>BookTitle/Author</b>                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------|
| R1         | Microbiology: A Laboratory Manual, 10th edition. James Cappuccino, Natalie Sherman. Pearson Higher education |
| R2         | Practical Microbiology, R C Dubey, D K Maheshwary                                                            |

### Online Resources

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                                                                     |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.           | <a href="https://microbenotes.com/pour-plate-technique-procedure-significance-advantages-limitations/">https://microbenotes.com/pour-plate-technique-procedure-significance-advantages-limitations/</a> |
| 2.           | <a href="https://microbiologyinfo.com/spread-plate-technique-principle-procedure-and-uses/">https://microbiologyinfo.com/spread-plate-technique-principle-procedure-and-uses/</a>                       |
| 3.           | <a href="https://www.ncbi.nlm.nih.gov/books/NBK562156/">https://www.ncbi.nlm.nih.gov/books/NBK562156/</a>                                                                                               |